Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 3 Quiz 1 - Forces and Dynamics

Mock Exam C · Form 1: Connecting ideas

Multiple Choice Questions: Select the best answer. (1 point each)

1. Equal forces at equal angles above or below horizontal act on a cart sliding right with kinetic friction and maintained contact. Which has the larger rightward acceleration?

(A) Angled upward

(B) Always equal

(C) Angled downward

(D) Mass alone decides

2. A 50 kg box initially at rest is pushed horizontally by 98 N. $\mu_s = 0.4$, $\mu_k = 0.25$. What is the actual friction magnitude? Use g = 9.8 m/s².

(A) -98 N

(B) 196 N

(C) 49 N

(D) 98 N

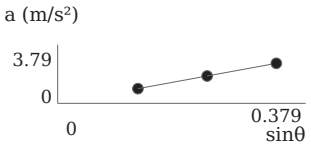
3. Release a block on a frictionless ramp and record downhill acceleration. Use g=10 m/s². sinθ -> a (m/s²): (0.1392, 1.392); (0.2592, 2.592); (0.3792, 3.792). Which interpretation of the experiment is valid?

(A) The slope is the first vertical reading.

(B) Changing block mass does not change this acceleration.

(C) Exchanging axes always preserves the numerical slope.

(D) Horizontal interval alone determines vertical change without a model.



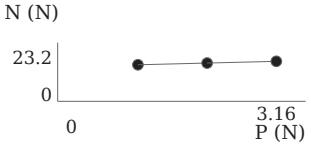
4. Press vertically down on a 2 kg block on a level floor. Use g=10 m/s². P (N) -> N (N): (1.16, 21.16); (2.16, 22.16); (3.16, 23.16). By what factor does the first-to-third change exceed the first-to-second change?

(A) 3.6

(B) 1.2

(C) 2.8

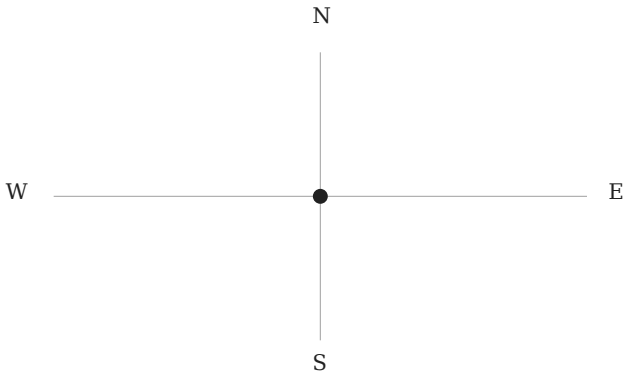
(D) 2



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. Two students push a cart of mass 12 kg on a frictionless horizontal floor. A exerts 260 N at 20° north of east. B exerts 360 N at 36° south of west.

a. Draw and label the actual forces acting in the plane of the floor on the dot. Draw whole vectors, not components. **(1 pt)**



b. Show net-force x and y components, then find acceleration magnitude and direction using compass words. **(3 pt)**

6. The same two surfaces slide against each other. Vary the normal force and measure friction magnitude. N (N) -> fk (N): (1.16, 0.464); (2.16, 0.864); (3.16, 1.264).

a. Build the model for sliding-friction measurements from physics and predict both slope and intercept. **(2 pt)**

b. Exchange the axes. Relate the new slope and its units to the original slope. **(2 pt)**

7. A mass-m cart slides right under force F at θ below horizontal. Its measured horizontal acceleration is a. Derive kinetic friction coefficient μ with zero vertical acceleration. **(3 pt)**